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P. Diwakaran

192424200

Slot-B

DSA0304

NLP

1. b) Passive production rules allow hierarchical sentence generation
2. a) Sub-Verb agreement.
3. a) Finite State Automaton
4. a) Morphological Analyzer.
5. b) Left recursion causing infinite expansion.
6. b) User, Predictor, Scanner and Completer operation to process partial input efficiently.
7. a) Regular expression matching.
8. a) Choose the parse tree with the maximum sub probability.
9. c) Finite state transducer + pointer scanner.
10. a) CFG + Features structure ✓
11. a) Remove all PP phrases $NP \rightarrow NP.PP$ ✗
12. a) The list of authorized employees were updated.
13. a) The verb violates its subcategorized frame. ✓
14. a) Incorrect probability ✓
15. b) Dynamic chart construction combining Predictor, scanner and completer ✓

35. c) No ✓
34. c) ✓
16. a) Number feature conflict between subject and verb. ✓
 17. a) Tree, because it has higher probability. ✓
 18. a) Highly recursive grammar with unrestricted production ✓
 19. d) Earley parsing handles ambiguity efficiently, while PCFG ranks ✓
 20. c) CFG Parsing → Subcategorization - Feature ✓
 21. c) Introduce a compound noun production $N \rightarrow \text{"living room"} N$
 22. a) Main subject noun phrase. ✓
 23. d) Verb subcategorization Frame ✗
 24. a) Shift reduce conflict. ✓
 25. a) Predictor c) completer. ✗
 26. b) Maximum likelihood estimation. ✗
 27. a) Tree A because it has the highest probability ✓
 28. a) context free grammar only. ✓
 29. b) Recursive top-down parser. ✓
 30. d) Highest PCFG rule Probability. ✗
 31. a) $S \rightarrow NP \rightarrow \text{Bot} \quad NVP \rightarrow \text{the robot} \quad VP \rightarrow VNP \rightarrow \text{delivers} \quad PP \quad A$ ✓
 32. a) The engineers develop the software ✓
 33. a) She gave the customer a refund. ✓
 34. d) Expand S into NP VP ✗
 35. b) Scanner. ✗
 36. b) Unification fails because feature value conflict. ✓
 37. a) The rule with the highest overall probability ✓

38. c) Recursive Descent Parsing. ✓
39. a) Parse A because it satisfies the constraints. ✓
40. b) Reject the parse because grammatical agreement. ✓
41. a) $VP \rightarrow VNP \propto$
42. a) PFT $\propto$
43. a) Predictor. ✓
44. a) Sentence accepted because PERSON matches. ✓
45. a) the maximum number of production rules.
46. a) CFG supports recursive non phrase generation using NP. ✓
47. a) Structural (Syntactic) ambiguity. ✓
48. a) Recursive grammar expansion. ✓
49. a) The verb "gave" require additional complements. ✓
50. a) $VNP PP \propto$ ✓